

Design and Implementation of a Low-Cost, Modular Wireless Security Assessment Platform using ESP32

Abstract—The proliferation of Internet of Things (IoT) devices has created an urgent need for accessible, portable, and robust security assessment tools. While commercial solutions such as the Flipper Zero and HackRF One exist, their high cost and unmitigated offensive capabilities present significant barriers to entry for educational institutions and developing regions. Furthermore, the lack of centralized governance in existing handheld tools poses legal and ethical risks when placed in the hands of students. This paper introduces “FZ-Lite,” a modular, low-cost wireless security assessment platform built upon the ESP32 microcontroller architecture. Unlike traditional offensive tools, FZ-Lite integrates a centralized Fleet Management Portal that enforces role-based access control (RBAC), audit logging, and a “receive-only by default” safety protocol. We demonstrate that by leveraging the ESP32’s native Wi-Fi and Bluetooth capabilities combined with modular RF front-ends, we can achieve 80% of the functional utility of industry-standard tools at approximately 15% of the cost. This architecture provides a scalable, legally compliant framework for democratizing offensive security education.

Index Terms—ESP32, Wireless Security, IoT Auditing, Red Teaming, Fleet Management, Cybersecurity Education, Secure Boot.

I. INTRODUCTION

The cybersecurity landscape has shifted dramatically toward the edge. With billions of IoT devices deployed globally, the attack surface has expanded beyond traditional enterprise networks into the Radio Frequency (RF) spectrum. To secure this environment, security professionals require portable tools capable of analyzing Sub-1GHz protocols, Wi-Fi frames, and Bluetooth Low Energy (BLE) advertisements.

Currently, the market for portable penetration testing hardware is dominated by devices based on the STM32 architecture, most notably the Flipper Zero. While highly capable, these devices present two critical challenges for academic and corporate training environments. First, the cost is prohibitive for mass deployment in developing economies. Second, these devices operate as autonomous “black boxes” with no oversight; once a device is issued to a student or junior analyst, there is no technical restriction on its use for illegal interference.

This paper proposes a novel architectural approach to portable security hardware. We present the design and implementation of a system that shifts the processing core from the STM32 to the more affordable and connectivity-rich ESP32-WROOM series. Recent research has highlighted the ESP32’s potential to democratize network security management due to its low cost and ubiquity [1]. Crucially, we introduce a “Fleet Management” paradigm, where the handheld device requires

cryptographic authorization from a central server to unlock active transmission capabilities.

Our contributions are as follows:

- 1) **Cost-Optimized Hardware Design:** A modular PCB design leveraging the ESP32 to reduce unit costs below \$35 USD.
- 2) **Governed Firmware Architecture:** A FreeRTOS-based system that defaults to passive analysis, requiring signed tokens to enable active injection.
- 3) **Centralized Audit Framework:** A Dockerized command-and-control portal that aggregates telemetry and audit logs for post-engagement analysis.

II. BACKGROUND AND RELATED WORK

The domain of portable wireless security assessment has evolved rapidly, transitioning from laptop-based software defined radios (SDRs) to compact, handheld multi-tools. This section reviews existing hardware platforms and highlights the specific gap in educational governance that FZ-Lite addresses.

A. Commercial and Enthusiast Hardware

The market leader in this category is the *Flipper Zero*, based on the STM32WB55 microcontroller. It consolidates Sub-1GHz, NFC, RFID, and GPIO capabilities into a consumer-friendly form factor. However, with a retail price exceeding \$169 USD, it remains inaccessible for bulk deployment in developing nations. Research by Salvador et al. demonstrated that Flipper Zero’s deauthentication signals can be easily fingerprinted, suggesting a need for more stealthy, governed architectures [3].

Another prominent tool is the *HackRF One*, a half-duplex SDR capable of operating from 1MHz to 6GHz. While it offers superior frequency agility compared to microcontroller-based tools, it requires an external host (laptop or smartphone) for operation, lacking the standalone portability required for field audits.

In the open-source community, the *Pwnagotchi* project utilizes a Raspberry Pi Zero W to capture WPA2 handshakes using Deep Reinforcement Learning. While effective as a passive capture tool, it operates autonomously without user intervention or audit logging [8].

B. ESP32-Based Security Implementations

The ESP32 SoC has gained widespread traction as a low-cost alternative for security research due to its integrated Wi-Fi and Bluetooth controllers. Beyond simple proof-of-concept

projects, the ecosystem has matured into highly sophisticated offensive firmware suites.

The most notable standard in this domain is the *ESP32 Marauder*, which provides a comprehensive suite for Wi-Fi and Bluetooth analysis. It is frequently deployed as a “backpack” companion for the Flipper Zero or as a standalone tool, enabling packet injection and PCAP logging on consumer hardware.

More recently, the *Bruce* firmware has emerged as a versatile, multi-hardware platform tailored for Red Team operations. Unlike single-vector tools, Bruce supports a wide array of peripherals including the M5Stack Cardputer, M5StickC, and T-Deck. It aggregates diverse offensive capabilities, ranging from beacon swarming and deauthentication to BLE spamming into a unified, menu-driven interface.

In the domain of covert operations, the *Nemo* firmware demonstrates the risk of miniaturization. Designed for the compact M5StickC and M5StickC Plus platforms, Nemo enables discrete execution of disruptive payloads, such as “AppleJuice” (BLE advertisement flooding) and “TV-B-Gone” (infrared shutdown), effectively turning a device the size of a USB drive into a potent offensive tool.

However, despite their technical efficacy, these existing ESP32 implementations suffer from critical limitations in an educational context:

- **Unmitigated Offensive Capability:** Firmwares like Bruce, Nemo, and Marauder are “offensive-by-default.” They do not distinguish between a student in a lab and an actor in the wild. There is no role-based access control (RBAC) to restrict features based on location or user authorization.
- **Lack of Centralized Governance:** They operate as autonomous silos. An instructor cannot remotely disable a student’s device or revoke transmission privileges if safety boundaries are violated.
- **Absence of Accountability:** There is no mechanism to enforce non-repudiation. If a device utilizing Nemo or Bruce is used to jam a legitimate network, there is no tamper-evident log linking the action to a specific user session or central fleet portal.

III. PROBLEM ANALYSIS

A. The STM32 vs. ESP32 Paradigm

Existing tools like the Flipper Zero utilize the STM32WB55 microcontroller. While efficient, this chip lacks native Wi-Fi capabilities, requiring external add-on boards (such as the Wi-Fi Devboard) for network testing. This increases complexity and cost.

The ESP32, conversely, integrates Wi-Fi (802.11 b/g/n) and Bluetooth/BLE natively. Historically, the ESP32 was viewed as power-inefficient for battery-powered tools. However, by implementing aggressive deep-sleep cycling and separating the RF frontend power rails, we demonstrate that the ESP32 is a viable candidate for handheld auditing tools.

B. The Governance Gap

Current tools are designed for individual hobbyists. In an institutional setting (e.g., a university lab or a corporate Red Team), there is no mechanism to prevent a user from jamming public frequencies or capturing handshakes outside authorized scope. This liability prevents many universities from adopting practical RF security curriculums.

IV. SYSTEM ARCHITECTURE

The FZ-Lite system is composed of three distinct layers: the Embedded Hardware Layer, the Firmware Logic Layer, and the Fleet Management Portal.

A. Hardware Layer

The core processing unit is the ESP32-WROOM-32D. We selected this module for its dual-core Xtensa LX6 architecture, capable of handling the heavy overhead of packet parsing while maintaining a responsive User Interface (UI).

- **Display:** 1.3-inch OLED (I2C) for real-time signal visualization.
- **Input:** 5-way tactile navigation switch for menu traversal.
- **RF Expansion:** A custom 8-pin GPIO header allows for “Hot-Swappable” modules. We successfully integrated a CC1101 module for Sub-1GHz analysis and an NRF24L01 for 2.4GHz proprietary protocols.
- **Storage:** MicroSD interface via SPI for logging PCAP (Packet Capture) files.

B. Firmware Architecture

The firmware is built on the Espressif IoT Development Framework (ESP-IDF) using FreeRTOS. To ensure system stability during high-load tasks (such as channel hopping), we pin the UI rendering task to Core 0 and the networking/packet sniffing tasks to Core 1.

A key innovation in our firmware is the `SAFE_MODE` flag. By default, this flag is hard-coded to `TRUE`.

```
void transmit_packet(uint8_t* buf, size_t len) {
    if (SYSTEM_STATE.safe_mode == true) {
        log_audit("Blocked_TX_attempt");
        ui_show_error("RX_ONLY_MODE");
        return;
    }
    esp_wifi_80211_tx(WIFI_IF_STA, buf, len, true);
    log_audit("Packet_Transmitted");
}
```

Listing 1. Simplified Safe Mode Logic

V. FLEET MANAGEMENT & GOVERNANCE

The distinguishing feature of FZ-Lite is its integration with a self-hosted web portal. The portal is developed using Python (Django/Flask) and containerized via Docker.

A. Token-Based Authorization

To enable active testing features (e.g., deauthentication frames or replay attacks), the device must connect to the portal. The instructor or team lead issues a time-limited JSON Web Token (JWT). The ESP32 verifies the signature of this token. If valid, the device unlocks active capabilities for the duration of the lab session.

B. Audit Trails

The device maintains a local log of all actions on the MicroSD card. Upon reconnection to the portal, these logs are uploaded to the central server. This ensures that even if a student attempts to hide their activity, the device syncs its history, creating a tamper-evident audit trail essential for legal compliance in educational settings.

VI. IMPLEMENTATION AND TESTING

The realization of the FZ-Lite architecture required solving significant engineering challenges, particularly in porting logic typically reserved for powerful SDRs or desktop CPUs onto a resource-constrained microcontroller. This section details the hardware fabrication, firmware optimization strategies, and the rigorous testing methodology used to validate the platform's efficacy and safety.

A. Hardware Fabrication and Integration

The physical implementation focuses on signal integrity and modularity. We utilized the ESP32-WROOM-32D as the primary MCU, clocked at 240 MHz.

- **RF Isolation:** A critical challenge in mixed-signal embedded design is noise coupling between the MCU's digital switching noise and the sensitive RF front-ends. We implemented a split-plane PCB design, separating the digital logic ground from the RF ground, bridged by a ferrite bead to suppress high-frequency noise.
- **Bus Contention Management:** The CC1101 (Sub-GHz), NRF24L01 (2.4GHz), and MicroSD card reader all share the VSPI bus. To prevent data corruption during simultaneous logging and scanning, we utilized a hardware Chip Select (CS) multiplexing strategy controlled by a dedicated GPIO interrupt handler.
- **Power Management:** We integrated a TP4056 charging controller with a low-dropout (LDO) regulator (HT7333) to step down the 3.7V LiPo battery to a stable 3.3V. Oscilloscope analysis confirmed that voltage ripple remained below 30mV peak-to-peak during active Wi-Fi transmission, ensuring stable operation of the RF modules.

B. Firmware Logic and Real-Time Optimization

The firmware is built upon the Espressif IoT Development Framework (ESP-IDF) v5.1, leveraging the FreeRTOS scheduler to manage concurrency.

1) *Dual-Core Asymmetric Processing:* The ESP32's Xtensa LX6 dual-core architecture was exploited to decouple user interface (UI) responsiveness from network packet processing.

- **Core 0 (Protocol Task):** Dedicates 100% of CPU cycles to the Wi-Fi/Bluetooth stack and promiscuous mode callbacks. This ensures that heavy packet ingress does not cause UI stutter.
- **Core 1 (Application Task):** Handles the OLED rendering (via I2C), button debouncing, and the Fleet Management cryptographic verification routines.

2) *High-Throughput Packet Buffering:* Directly writing every captured packet to the SD card creates a bottleneck due to flash write latency (typically 10-20ms). In high-density environments (e.g., a lecture hall with 200+ devices), this latency leads to packet loss. We optimized the promiscuous mode callback to handle high-throughput ingress, a challenge noted in previous performance analyses of low-power IoT sniffers [7]. We implemented a **Circular Ring Buffer** using the ESP32's DMA (Direct Memory Access). Incoming packets are pushed to a 16KB DMA buffer in RAM. A low-priority background task creates batched writes to the SD card only when the buffer reaches 75% capacity.

$$L_{write} = \frac{S_{buffer}}{R_{ingress}} \times \eta \quad (1)$$

Where L_{write} is the write latency, S_{buffer} is buffer size, and η is the SPI throughput efficiency. This architecture reduced packet drop rates from 14.3% to less than 1.2% in our stress tests.

C. Fleet Portal Integration

The device communicates with the Dockerized management portal via HTTPS using mutual authentication. Upon boot, the device generates a unique hardware fingerprint (derived from the factory-fused MAC address) and requests a configuration manifest. The portal responds with a signed JSON payload containing the *Rules of Engagement (RoE)*:

- `allowed_channels`: [1, 6, 11]
- `feature_flags`: { "deauth": false, "pcap": true }
- `session_timeout`: 3600 (seconds)

The firmware parses this JSON and dynamically locks/unlocks menu items in the UI. This proves that the device's behavior is software-defined by the instructor, not the student.

D. Experimental Validation

We conducted three distinct testing scenarios to benchmark FZ-Lite against the industry-standard Flipper Zero and a generic Alfa AWUS036NHA adapter.

1) *Scenario A: Packet Capture Sensitivity:* We placed a target transmitter emitting Beacon frames at -80dBm fixed power. We measured the Receive Signal Strength Indicator (RSSI) and total frame count over a 60-second interval.

TABLE I
SENSITIVITY COMPARISON (AVERAGE OF 10 RUNS)

Device	Avg RSSI	Packets Captured	Loss Rate
Alfa AWUS036NHA	-78 dBm	5984	0.2%
FZ-Lite (ESP32)	-82 dBm	5890	1.8%
Flipper Zero (WiFi Dev)	-81 dBm	5910	1.5%

The results (Table I) indicate that while the dedicated high-power Alfa card performs best, the FZ-Lite is statistically comparable to the much more expensive Flipper Zero in passive capture tasks.

2) *Scenario B: Battery Endurance Profiling*: Portability is a key requirement. We profiled the power consumption of FZ-Lite using a Nordic Power Profiler Kit II (PPK2).

- **Idle (Menu)**: 45mA
- **Passive Scanning (WiFi)**: 110mA
- **Active Attack (Deauth)**: 180mA (Peak)

With a standard 1500mAh LiPo battery, FZ-Lite achieves approximately 9 hours of continuous passive scanning functionality. In contrast, the Flipper Zero (with its proprietary sub-GHz efficient modes) achieves 20+ hours, highlighting the trade-off inherent in using the more power-hungry Wi-Fi native ESP32.

3) *Scenario C: The "Classroom Chaos" Stress Test*: To simulate a realistic educational environment, 30 FZ-Lite units were activated simultaneously in a 40x40 ft lab. The Fleet Portal successfully handled the concurrent "check-in" requests with a mean response time of 240ms. Crucially, when the instructor initiated a global "STOP" command via the portal, 100% of the devices ceased sniffing operations within 3 seconds of their next heartbeat interval, validating the centralized governance mechanism.

VII. THREAT MODEL AND PLATFORM SECURITY

In an educational environment, the threat landscape is unique: the primary adversaries are the users (students) themselves, who possess physical access to the device and may be motivated to bypass governance controls for curiosity or malicious intent. Therefore, FZ-Lite adopts a "Zero Trust" approach to hardware trust.

A. Adversary Capabilities

We define the adversary model ($\mathcal{A}$) with the following capabilities:

- **Physical Access**: $\mathcal{A}$ has unrestricted physical access to the FZ-Lite device, including the USB-UART bridge and JTAG interface.
- **Firmware Manipulation**: $\mathcal{A}$ attempts to flash unmanaged, offensive firmware (e.g., generic ESP32 Marauder binaries) to bypass the fleet audit trail.
- **Secret Extraction**: $\mathcal{A}$ attempts to dump the SPI flash memory to extract the Fleet Authorization Token or the institutional Wi-Fi credentials.

B. Hardware Root of Trust

To mitigate firmware replacement attacks, FZ-Lite leverages the ESP32's **Secure Boot V2** architecture. The bootloader verifies the RSA-3072 signature of the application image against a digest of the public key stored in the hardware eFuses.

$$\text{Verify}(\text{App}_{img}, K_{pub}) \rightarrow \{\text{Boot} \mid \text{Halt}\} \quad (2)$$

If a student attempts to flash a binary not signed by the instructor's private key (K_{priv}), the bootloader detects the signature mismatch and permanently halts the boot process, rendering the device inert. This ensures that only governed, audit-enabled firmware can execute on the hardware.

C. Data-at-Rest Protection

Physical extraction of stored secrets (e.g., previous audit logs or cached configurations) is mitigated via **Flash Encryption**. We utilize the XTS-AES-256 algorithm, where the 256-bit encryption key is generated internally via the hardware Random Number Generator (RNG) and burned into the BLOCK_KEY eFuse. Crucially, we set the FLASH_CRYPT_CNT bit to max, effectively disabling the ability to disable encryption later. Even if an adversary desolders the SPI flash chip and reads it with an external programmer, the data appears as high-entropy noise.

D. Ephemeral Token Storage

The JSON Web Token (JWT) used to authorize offensive capabilities is classified as a critical asset. To prevent token theft, we implement a strict **Volatile Storage Policy**. The JWT is never written to the non-volatile storage (NVS) partition. It resides solely in the ESP32's internal SRAM. Consequently, any power loss or hardware reset (Watchdog Timer) instantaneously clears the authorization state. This "fail-safe" design ensures that a lost or stolen device inherently reverts to a safe, unauthorized state.

VIII. ETHICAL CONSIDERATIONS & COMPLIANCE

The development of dual-use technology—tools capable of both defense and offense—requires a rigorous ethical framework. FZ-Lite is architected to enforce the principle of **Non-Repudiation**. Our "Safety-by-Design" approach aligns with the ethical training principles proposed by Heiding and Lagerström, shifting the focus from exploitation to responsible auditing [5].

A. Legal Compliance Framework

In the context of the *Pakistan Wireless Telegraphy Act* and international ITU regulations, unauthorized transmission on ISM bands can be classified as illegal interference. FZ-Lite's default firmware is legally compliant by design:

- **Passive-by-Default**: The radio hardware is initialized in *Promiscuous Mode* with TX amplifiers disabled.
- **Geofencing (Future Work)**: We propose integrating Wi-Fi BSSID geolocation to restrict active testing features strictly to laboratory coordinates.

B. Responsible Education

By mandating the use of the Fleet Portal, we shift the psychological model of the student from "hacker" to "auditor." The existence of the immutable audit log serves as a digital Panopticon, discouraging misuse not through physical inability, but through guaranteed accountability.

IX. COST ANALYSIS AND ECONOMIC FEASIBILITY

The primary barrier to the widespread adoption of hardware-based security training in developing economies is the prohibitive cost of commercial tools. The Flipper Zero, while revolutionary in its form factor, retails for approximately \$169 USD. In the context of an educational institution in Pakistan,

outfitting a single lab of 50 students with Flipper Zero units would require a capital expenditure exceeding \$8,450 USD (excluding import duties and shipping).

FZ-Lite fundamentally disrupts this economic model by shifting the architecture from proprietary, high-markup components to commodity, mass-market silicon.

A. Comparative Bill of Materials (BOM)

We performed a granular cost breakdown of the FZ-Lite prototype against the retail pricing of the Flipper Zero. The costs for FZ-Lite are derived from local electronic markets in Pakistan (e.g., Hall Road, Lahore) and standard AliExpress bulk pricing.

TABLE II
COST EFFICIENCY ANALYSIS: FZ-LITE VS. INDUSTRY STANDARD

Subsystem	Specification	Flipper Zero (Ref)	FZ-Lite (Proposed)
Compute Core <i>Est. Cost</i>	Primary MCU	STM32WB55 (Cortex-M4) ≈\$14.50	ESP32-WROOM-32D ≈\$3.80
RF Capabilities <i>Est. Cost</i>	Sub-1GHz/2.4GHz	Integrated CC1101 Included in Mainboard	Modular (CC1101/NRF24) ≈\$5.50 (Modules)
Wi-Fi Audit <i>Est. Cost</i>	802.11 b/g/n	Requires External PCB ≈\$40.00 (Devboard)	Native SOC Feature \$0.00 (Integrated)
PCB Fab <i>Est. Cost</i>	Stackup & Finish	4-Layer HDI / ENIG High Complexity	2-Layer FR4 / HASL Low Complexity
Total Cost	Acquisition Price	≈\$209.00 USD*	≈\$32.00 USD

*Includes Flipper Zero Unit (\$169) + Wi-Fi Devboard (\$40) for feature parity.

B. The Democratization Factor

As illustrated in Table II, the FZ-Lite achieves a **83.4% reduction in hardware costs** compared to the market leader. Ideally, the native integration of Wi-Fi in the ESP32 eliminates the need for the "Wi-Fi Devboard" (a \$40 accessory for the Flipper Zero), further widening the value gap.

This price point (\$28 USD / ≈7,800 PKR) enables a "1:1 Student-to-Device Ratio." Instead of a shared lab resource that students check out for an hour, the FZ-Lite is affordable enough to be included in student tuition as a personal, take-home learning kit.

C. Supply Chain Resilience

A critical feasibility factor for the Pakistani market is component availability. The STM32WB55 used in the Flipper Zero faced significant shortages during the 2021-2023 global chip crisis. In contrast, the ESP32-WROOM and the discrete RF modules (CC1101, NRF24) are "jellybean parts"—ubiquitous components available in massive quantities from multiple vendors. This ensures that the FZ-Lite can be manufactured locally with minimal supply chain risk, supporting our goal of indigenous production.

X. CONCLUSION

This paper presented the design of FZ-Lite, an ESP32-based security toolkit that challenges the assumption that effective Red Teaming hardware must be expensive or proprietary. By

decoupling the hardware from the high-cost STM32 ecosystem and enforcing software-defined governance via a Fleet Portal, we have created a solution tailored for the next generation of cybersecurity professionals.

Future work includes the development of a "Purple Team" mode, where the device simulates specific IoC (Indicators of Compromise) patterns to test Blue Team detection capabilities automatically. Future iterations could also integrate Reinforcement Learning for automated handshake capture, similar to the methodology used in the Pwnagotchi project [8].

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